Science, connected

A current overview | Checked 26 October 2026

PURPOSE

MeatCODE is the Meat Collaborative Open Development Ecosystem, a GFI Israel initiative turning scattered meaty flavor and aroma science into a searchable knowledge hub for alternative-protein R&D. The prototype is under validation; Lior Teper builds it and Daniel Dikovsky leads the GFI IL SciTech work.

PLATFORM

Oracle scientific Q&A, literature exploration, molecule and expert discovery, and research views. GC-MS interpretation, simulation and experimental planning are intended journeys with varying implementation and validation status.

WHAT EXISTS NOW

818 sources · 799 molecule rows · 3,129 experts · 45 claims
Neon production: 49 base tables + 16 views. Dev: 56 base tables + 16 views. Both have 3,650 molecule-property values and 23 paper-molecule links; experiments and processes are empty.

GRAPH AND RETRIEVAL

Local LightRAG: 5,101 nodes / 6,240 edges. The frozen Neon dev projection is a different snapshot: 3 documents, 3,418 entities, 4,164 relations and 7,582 vector(1024) embeddings. Extracted graph data still needs scientific curation.

HOW IT WORKS

Ingest: PDF > OCR if needed > GROBID TEI > LightRAG > Neon dev / pgvector.
Answer: Oracle > server > evidence retrieval > Claude > streamed cited response. Graph serving is dev-only in local code. Independent M2 validation is a target design.

CURRENT CHECKS AND NEXT WORK

Production reports 6394c039749a, main, claude-sonnet-4-6, password protected. Deployed retrieval logic and the earlier Oracle credential issue were not retested. Priorities: curate evidence, strengthen scientific links, and run the RAG-versus-direct-LLM benchmark.

WHERE TO GO

github.com/Bokipr0/meatCODE (private) · Live platform
Code: app/ · server/ · db/ · pipeline/ · analysis/
Context: platform_docs/ · orchestration/ · Release Center/
Pack entry point: READ_ME_FIRST.md; complete columns and relationships in schema/SCHEMA_REFERENCE.md.